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Research and Applications

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Leveraging deep learning to detect stance in Spanish tweets on COVID-19 vaccination

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Abstract

Objectives: The automatic detection of stance on social media is an important task for public health applications, especially in the context of health crises. Unfortunately, existing models are typically trained on English corpora. Considering the benefits of extending research to other widely spoken languages, the goal of this study is to develop stance detection models for social media posts in Spanish.

Materials and Methods: A corpus of 6170 tweets about COVID-19 vaccination, posted between March 1, 2020 and January 4, 2022, was manually annotated by native speakers. Traditional predictive models were compared with deep learning models to ascertain a baseline performance for the detection of stance in Spanish tweets. The evaluation focused on the ability of multilingual and language-specific embeddings to contextualize the topic of those short texts adequately.

Results: The BERT-Multi+BiLSTM combination yielded the best results (macroaveraged F1 and Matthews correlation coefficient scores of 0.86 and 0.79, respectively; interpolated area under the receiver operating curve [AUC] of 0.95 for tweets against vaccination and 0.85 in favor of vaccination and a score of 0.97 for tweets containing no stance information), closely followed by the BETO+BiLSTM and RoBERTa BNE-LSTM Spanish models and the term frequency-inverse document frequency+SVM model (average AUC decrease of 0.01). The main differentiating factor among these models was the ability to predict tweets against vaccination.

Discussion: The BERT Multi+BILSTM model outperformed the other models in terms of per class prediction capacity. The main assumption is that language-specific embeddings do not outperform multilingual embeddings or TF-IDF features because of the context of the topic. The inherent context of BERT or RoBERTa embeddings is general. So, these embeddings are not familiar with the slang commonly used on Twitter and, more specifically, during the pandemic.

Conclusion: The best performing model detects tweet stance with performance high enough to ensure its usefulness for public health applications, namely awareness campaigns, misinformation detection and other early intervention and prevention actions seeking to improve an individual's well-being based on autoreported experiences and opinions. The dataset and code of the study are available on GitHub.

Lay Summary

Stance detection can be defined as the process of automatically determining the author's viewpoint and judgment toward a predetermined topic. Health-related applications can benefit greatly from automatic stance detection given the polarization affecting certain topics (eg, vaccination) and the growing number of social media users looking for and discussing health information. Notably, social media communication during health crises, namely the COVID-19 pandemic, has motivated further research on detecting the opinions and concerns of individuals during such stressful, constantly changing periods. Unfortunately, existing models are typically trained on English corpora. This study investigates stance detection on social media posts in Spanish. Several traditional and deep learning methods are evaluated. Posting contents, namely slang, jargon, and topic-specific terms, are represented using multilingual and language-specific embeddings. Results show that people's stance can be monitored with pretrained multilingual resources. The best performing model detects tweet stance with sufficient performance to ensure its usefulness for public health applications, namely awareness campaigns, misinformation detection, and other early intervention and prevention actions seeking to improve an individual's well-being. The generalization to other biomedical domains will require the inclusion of topic-specific terminology to better capture social contexts. The dataset and code of the study are available on GitHub.

Key words: stance; social media; Spanish; transformer embeddings; deep learning.

Introduction

The task of stance detection can be defined as the process of automatically determining the author's viewpoint and judgment toward a predetermined topic.[1] Therefore, stance detection implies determining the agreement or disagreement of the text regarding the target, usually as being in favor, against, or neither.[2,3] The target may not be mentioned in the text and stance is not always directly associated with the polarity of the text.[4] The aim of the task is thus to infer stance based on available information, that is, linguistic style and metrics, social connections, and source identity.

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This task is considered invaluable to many applications of Natural Language Processing and Text Mining, such as political debates and prediction of election/referendum results,[5] rumor classification,[5] and misinformation detection.[2] Health-related applications are included in the above research but pose problems of their own, given the polarization affecting certain topics (namely vaccination[6–8]) the growing number of social media users looking for and discussing health information,[9,10] and the mixing of social jargon (including nonvocabulary words, abbreviations, irony, and sarcasm) with medical terminology.[11–13] In particular, the relevance of social media communication during health crises, namely the COVID-19 pandemic,[14–16] has motivated further attention to the task and urged better mechanisms of grasping the mood, opinions, and concerns of individuals during such stressful, constantly changing periods.[17,18]

Community challenges are boosting social media research. The Social Media Mining for Health Research and Applications (#SMM4H) workshops are prominent representatives of these efforts, as they introduce tasks relevant to health applications and extend English research to other widely spoken languages. The #SMM4H 2020 included tasks for detecting adverse effects of medications in French and Russian tweets, characterizing chatter related to prescription medication abuse, and detecting self-reports of birth defect pregnancy outcomes.[19] The #SMM4H 2021 hosted a shared task to identify occupations mentioned in Spanish by Twitter users.[20] In 2022 and 2023, #SMM4H proposed shared tasks for classifying tweets on self-reported COVID-19 symptoms and detecting disease mentions in Spanish tweets.[21,22]

The "SemEval-2016 Task 6: Detecting Stance in Tweets" was the first community challenge addressing the determination of the stance of social posts.[23] The topics, including politics, religion, climate change, and the legalization of abortion, varied. The following SemEval workshop invested in a shared task to help create high-quality annotated datasets for automatic stance detection.[24] Then, #SMM4H 2022 proposed the first health-oriented task, that is, the detection of stances and premises in tweets about health mandates related to COVID-19 (in English).[25]

The limited availability of stance detectors for languages other than English led to new community challenges. For example, the determination of political stances on Spanish and Catalan tweets, that is, the "Stance and Gender Detection in Tweets on Catalan Independence"[26] and the "MultiModal Stance Detection in tweets on Catalan #1Oct Referendum (MultiStanceCat)."[27] More recently, a stance detection-shared task has also been performed for tweets in Italian,[28] and the VaxxStance-shared task at IberLEF 2021 proposed multilingual and cross-lingual tasks over Spanish and Basque tweets referring to vaccines.[29] These shared tasks favored the emergence of various machine learning approaches. Most notably, both classical models and deep learning architectures have been successfully trained to extract knowledge about the stance of social posts. Some studies have also investigated the portability of existing stance detection tools to multilingual settings.[30]

These successful tools and applications motivated this work, which investigated the performance of deep learning models for stance detection in Spanish social posts. The case study involves COVID-19 vaccine discussions on Twitter, which is practical and technically relevant considering that vaccination is susceptible to high polarization and confirmation bias and is strongly affected by misinformation. Therefore, the present study focused on the potential of deep learning architectures in grasping the limited context offered by Spanish tweets. To the best of the authors' knowledge, little Spanish-dedicated research exists aside from the works from the teams participating in the VaxxStance-shared task.[29] Therefore, the objectives of this work are 2-fold. First, the study of the performance differences between baseline classifiers and deep learning models was applied to Spanish tweets. Second, the capacity of multilingual and language-specific embeddings to represent the contexts of such short texts is explored to understand the impact they have on the downstream performance of the model. The generated results and existing limitations are discussed in detail, and the manually annotated corpus of stance in COVID-19 Spanish tweets and the best-performing model are made publicly available to promote further community research (https://github.com/gbgonzalez/stance_detection_sp_vaccines).

Materials and methods

Data collection and annotation

Data were collected between March 1, 2020 and January 4, 2022. The Tweepy Python library, which complies with the public Twitter REST API, supported such retrieval. The search queries included common mentions of the disease (eg, #coronavirus, #coronavirusespaña, #coronavirusmadrid, #COVID19, "coronavirus," "covid-19"), general vaccine-related terms (eg, "vacuna coronavirus," "vacuna," "inyec-," and "incocul-") and references to the vaccines with the most media exposure (eg, "Pfizer," "Moderna," "biontech," "AstraZeneca," "vacunaoxford," "oxford," "Johnson & Johnson," "Novavax," and "Sputnik"). The API also supported the filtering of search results by language. Only original tweets containing mentions of COVID-19 vaccines were retained; that is, retweets and duplicate tweets were eliminated.

The annotation guidelines were based on those that guided the production of the first English stance dataset,[31] and labeling was 3-fold, that is, "Against," "In favor," and "None." From all the retrieved tweets, a set of 6170 tweets was randomly selected, preserving the time distribution to avoid content bias. Each tweet was independently annotated by 3 native Spanish speakers. See Supplementary Material S1 for further details on the annotation guidelines and interannotator agreement.

A stratified 90:10 split was applied to obtain the training and test sets for the machine learning experiments. Table 1 shows the distribution of tweets and users per stance label and evaluation set. "None" was the predominant label and had the largest number of unique users. Conversely, tweets against COVID-19 vaccination were less frequent, representing the stance of a minor fraction of contributing users. The dehydrated training and test datasets can be found in Supplementary Material S2.

Tweet preprocessing

First, all identifiable information, such as email addresses, URLs, and mentions, were eliminated. The texts were converted to lowercase. Line breaks and white space sequences, repeated punctuation symbols, accents, and non-ASCII characters were removed. Hashtag marks were removed, leaving the associated text as is. For the generation of term

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Table 1. Manually annotated Spanish tweet corpus on the COVID-19 vaccination stance.

Train	Tweets	In favor	1498
		Against	926
		None	3129
		Total	5553
Test	Tweets	In favor	145
		Against	119
		None	353
		Total	617

frequency-inverse document frequency (TF-IDF) features, punctuation marks and stopwords were also removed, and the text was lemmatized. Conversely, the tweets were tokenized and converted into the evaluated embeddings.

Stance detection models

The performance of classical machine learning algorithms, that is, support vector machines (SVMs), complement Naïve Bayes (CNB), and random forest (RF), was compared with the performance achieved by 3 deep learning architectures: a linear neural network (LNN),[32] bidirectional long short-term memory (BiLSTM),[33] and an adaptation of the convolutional neural network (CNN) architecture for sentence classification.[34] The methods were chosen based on the performance they obtained on previous related research.

Support vector machines

Support vector machine is based on the search for a hyperplane that best separates the possible classes of data points.[35,36] The implementation chosen was the C-Support Vector Classification because it supports multiclass prediction.[37] Support vector machine models have been successfully applied to stance detection toward vaccination as well as other health topics.[31,38–42]

Complement Naïve Bayes

Complement Naïve Bayes is an implementation of the complementary Naive Bayes algorithm.[43] This modeling algorithm was explored because it usually works well on imbalanced datasets[44] and has yielded good results in monitoring the tweet stance on the European Union (EU) digital COVID-19 certificate[45] and Twitter stance detection toward the Job Creation Bill.[46]

Random Forest

The RF method belongs to the ensemble methods group.[47] Previous applications include hate speech detection on social media[48] and general stance detection in tweets.[45]

Linear neural network

Linear neural networks are the simplest architecture used in deep learning methods. In particular, LNNs are very commonly evaluated when applying BERT embeddings. Linear neural networks have been examined in different text understanding and classification tasks,[49,50] including stance detection.[51,52]

Bidirectional long short-term memory

A bidirectional long short-term memory network is an extension of the traditional single direction (forward) long short-term memory (LSTM) architecture that trains 2 LSTMs simultaneously to capture both forward and backward information of the input sequence.[53–57] Recent works include stance detection on Vietnamese and Italian social media.[58,59]

Convolutional neural network

A convolutional neural network (KimCNN) is a type of multilayer neural network that has been successful in several text classification tasks, including stance detection.[34] Among other applications, convolutional architectures have been used to detect stances in tweets using character and word-level CNN[60] and via knowledge distillation.[56]

Input features

Classical machine learning algorithms used as input features the TF-IDF scores.[61] Furthermore, experiments investigated a variety of state-of-the-art embeddings, such as BERT (BERT-Multi)[62] and XML-RoBERTa multilingual[63] embeddings, and the BETO,[64] BERTIN,[65] and RoBERTa BNE[66] Spanish pretrained embeddings.

Multilingual models are models that are trained simultaneously using data from several languages. BERT-Multi is a version of the BERT model trained in English and 104 different languages.[62] XML-RoBERTa is a competitor multilingual model that has obtained state-of-the-art performance on cross-lingual classification, sequence labeling, and question answering.[60,67] Multilingual embeddings have yielded promising results on political multilingual stance detection on Twitter,[68] stance detection in Italian tweets,[57] and tweet emotion prediction.[69]

The release of language-specific embeddings is expected to improve results but stance-specific comparison with multilingual representations is still underreported. The Spanish-BERT (BETO)[61] is based on the Whole Word Masking variant of BERT.[70] BERTIN,[65] that is, a series of transformer-based models trained on the Spanish portion of the mC4 dataset, and RoBERTa BNE[65] are RoBERTa[71] implementations for Spanish. BETO has already been used to detect stance in Spanish texts.[72] XML-RoBERTa and RoBERTa BNE have been evaluated on sentence classification tasks,[72,73] but have not been applied to stance detection in short texts.

Evaluation and experimental setup

All the models were trained with 10-fold cross-validation and evaluated with an independent test that represents ≈10% of the total dataset (ie, 617 tweets). The selection of the hyperparameters was based on the highest macroaveraged F1 score through cross-validation for each model.[74]

The grid search approach was applied to explore parameter combinations as described below. For SVM, $C \in \{0.1, 1, 10, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000\}$, kernel $\in \{\text{rbf, poly, linear, sigmoid}\}$, and $\gamma \in \{1, 0.1, 0.01, 0.001, 0.0001\}$. For CNB, $\alpha \in \{\text{logspace}(0, -9, \text{num} = 100)\}$. For RF, estimators $\in \{100, 300, 500, 8000, 1200\}$, maximum depth $\in \{5, 8, 15, 25\}$, minimum sample split $\in \{2, 5, 10, 15, 100\}$, and minimum sample leaf $\in \{1, 2, 5, 10\}$.

Regarding the neural network architectures, parameter search was as follows: dropout $d \in \{0.3\}$, Adam $\varepsilon \in \{0.9, 0.9999, 1e-8\}$, learning rate $\alpha \in \{2e-5, 3e-5, 5e-5\}$, epochs $n \in \{2, 3, 4\}$ in case of pretrained embedding and $n \in \{1-30\}$ in case of TF-IDF. Two batch sizes of 16 and 32 were used. Exception made for the models

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that used RoBERTa embeddings, which also considered a 64 batch size. The optimal hyperparameters are described in Supplementary Material S3.

All classifiers are evaluated against the same test partition after separate 10-fold cross-validation. Furthermore, a nested cross-validation approach was applied to estimate the generalization performance of the traditional classifiers (implemented as described by Kolchinsky et al.[75]) The performance of the classifiers was evaluated based on the F1 score, the area under the receiver operating curve (AUC), and the area under the interpolated precision/recall curve (iAUC),[76] and the Matthews correlation coefficient (MCC),[77] which is particularly well suited for the unbalanced scenarios of the present corpus.[78] While the MCC and F1 score provided insight into the performance of the different classifiers, the AUC and iAUC measures facilitated distinguishing performance differences per class, that is, in favor or against vaccination.

Results

Table 2 presents the performance obtained for the explored feature sets and multiclass models. For the traditional classifiers, the observed performance differences between pairs of configurations were tested for statistical significance via a nonparametric paired-sample permutation test. The *P*-value was computed as the probability of observing a difference under the null hypothesis distribution equal to or greater than the actual difference. Next, the performance of the deep learning methods was checked to determine whether they were significantly beyond the expected performance variance of traditional classifiers. The utility of deep learning is further established through the analysis of training-validation loss. Finally, an ablation study supports the study of the impact of various components on the performance of the best performing models, that is, the TF-IDF+SVM, BERTmulti+BiLSTM, BETO+BiLSTM, and RoBERTa BNE+LSTM models.

Table 2 shows that the BiLSTM model using BERT multilingual embedding yielded the best performance on average and per class basis. The BiLSTM models using BETO and XML-RoBERTa BNE embeddings have similar average AUC and iAUC scores (ie, a decrease of 0.01). The differences are in class-specific performance, namely in detecting stances against vaccination. The SVM model (which uses TF-IDF features) is the best performing model among traditional classifiers, and most notably, the performance differences among them are statistically significant ($P < .05$).

Since the performance of TF-IDF+SVM is not too different from that of the best deep learning models, one may prefer such a simpler classification model for explainability and feature identification. Moreover, this is a good choice considering that deep learning requires very large training corpora and this is not easily achieved.

The BERT-Multi+BiLSTM and the RoBERTa BNE+BiLSTM models showed the highest ability to discriminate tweets in favor of vaccination (Figure 1). The BERT-Multi+BiLSTM model was the best at detecting tweets showing a stance against COVID-19 vaccination. All the models exhibiting high performance across the metrics showed similar abilities to detect postings with no stance. The models using XLM-RoBERTa and BERTIN embeddings showed poorer performance for all categories (iAUC). Overall, the KimCNN was the worst-performing architecture for detecting stances against vaccination.

Training and validation loss analysis

After each training epoch, each deep learning model was evaluated on the training dataset (ie, to assess how well the model was learning) and on a hold-out validation set to check its

Table 2. Prediction performance is reported per class label and on average.

					$iAUC_{macro}$			
Embedding	**Classifier**	**F1 $score_{macro}$**	**MCC**	**AUC**	**AVG**	**Against**	**None**	**In favor**
TF-IDF	SVM	**0.86**	**0.79**	**0.95**	**0.90**	**0.91**	**0.96**	**0.82**
	CNB	0.85	0.76	0.95	0.90	0.91	0.96	0.83
	RF	0.69	0.58	0.90	0.83	0.84	0.92	0.84
	LNN	0.80	0.66	0.93	0.86	0.82	0.95	0.80
	BiLSTM	0.82	0.69	0.94	0.87	0.84	0.96	0.82
	KimCNN	0.42	0.00	0.50	0.67	0.60	0.79	0.62
BERT-Multi	LNN	0.86	0.79	0.94	0.88	0.94	0.93	0.78
	BiLSTM	**0.86**	**0.79**	**0.96**	**0.92**	**0.95**	**0.97**	**0.85**
	KimCNN	0.50	0.48	0.92	0.84	0.80	0.94	0.77
BETO	LNN	0.83	0.84	0.92	0.85	0.87	0.93	0.74
	BiLSTM	**0.85**	**0.85**	**0.96**	**0.91**	**0.92**	**0.97**	**0.84**
	KimCNN	0.77	0.72	0.93	0.86	0.84	0.95	0.80
BERTIN (RoBERTa Spanish)	LNN	0.83	0.75	0.90	0.81	0.81	0.90	0.72
	BiLSTM	0.84	0.76	0.95	0.90	0.92	0.96	0.83
	KimCNN	0.70	0.58	0.90	0.79	0.71	0.93	0.75
RoBERTa BNE (RoBERTa Spanish)	LNN	0.84	0.76	0.92	0.87	0.87	0.91	0.81
	BiLSTM	**0.84**	**0.76**	**0.96**	**0.90**	**0.89**	**0.97**	**0.85**
	KimCNN	0.78	0.69	0.94	0.87	0.84	0.95	0.82
XLM-RoBERTa (multilingual)	LNN	0.83	0.74	0.92	0.85	0.85	0.93	0.76
	BiLSTM	0.83	0.75	0.94	0.87	0.85	0.95	0.79
	KimCNN	0.73	0.63	0.93	0.85	0.82	0.94	0.79

Bold denotes the best performance across metrics.
Abbreviations: AUC, area under the receiver operating curve; BiLSTM, bidirectional long short-term memory; CNB, complement Naïve Bayes; iAUC, area under the interpolated precision/recall curve; LNN, linear neural network; MCC, Matthews correlation coefficient; RF, random forest; SVM, support vector machine; TF-IDF, term frequency-inverse document frequency.

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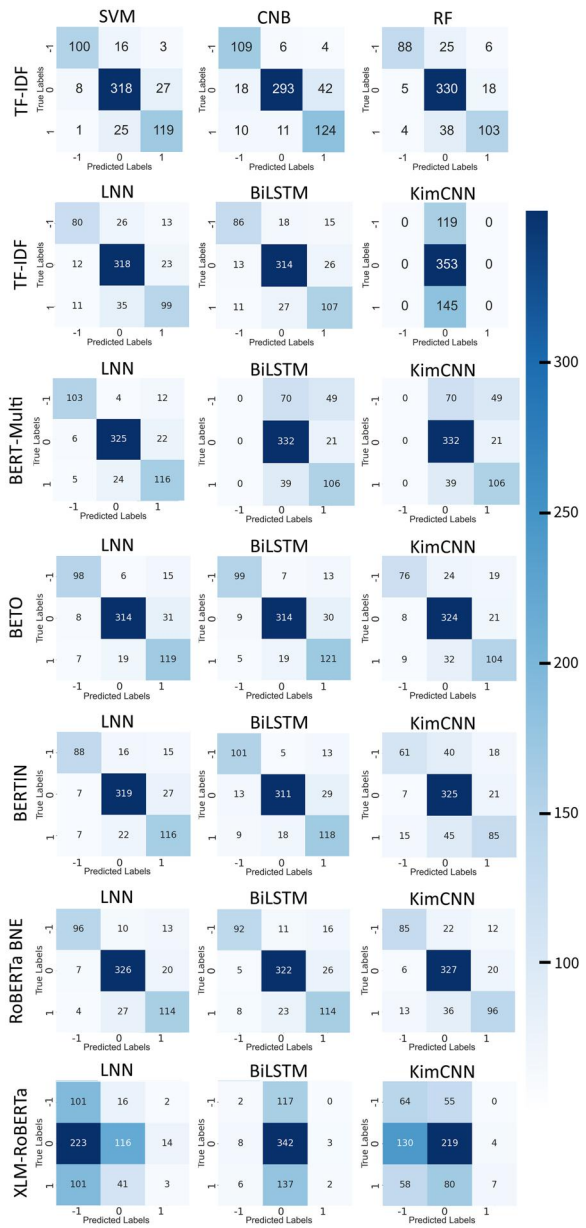

Figure 1. Confusion matrix for the test results. A row represents the actual tweets stance (−1 as against, 0 as none, and 1 as in favor), whereas a column represents the predicted tweets stance. For example, in the matrix of RoBERTa BNE+BiLSTM, the false negative predictions of no stance tweets (a total of 31) were mostly annotated as tweets in favor of vaccination (ie, 26). In the same way, the diagonal elements represent correct predictions.

performance on previously unseen data (ie, to assess how well the model was generalizing). As illustrated in Figure 2, the training loss for most of the models using embeddings steadily decreases with more epochs and converges to zero when the early stopping is reached. For most models, the validation loss also decreases with more epochs. However, BiLSTM models show a validation loss greater than the training loss after epoch 2, which is generally indicative of overfitting on the training examples.

Among the deep learning classifiers, the KimCNN models show the highest loss at the beginning and smaller training and validation loss improvements throughout training. Similarly, models using TF-IDF scores are unable to decrease training or validation loss further than 0.4.

Ablation study

Different ablation study experiments were performed for the best-performing models. With respect to the TF-IDF+SVM model, the study included different combinations of features, that is, unigrams, bigrams, and trigrams (Table 3).

The TFIDF+SVM model achieves the best performance when unigrams and a vector size of 1024 are used. The combined use of unigrams and bigrams and uni-, bi-, and trigrams also yields very high results. However, the exclusive use of unigrams shows slightly higher performance in tweets with a stance in favor and against vaccines.

Experiments on the best deep learning models, that is, BERT-Multi+BiLSTM, BETO+BiLSTM, and RoBERTa BNE+BiLSTM, were performed to study the impact of changing different parts of each of the architectures. This included a different number of neurons in the hidden layer and the use of different loss functions, that is, negative log-likelihood loss (NLLLoss) or cross-entropy loss (CrossEntropyLoss), and optimizers, that is, AdamW, AdaGrad, and stochastic gradient descent (SGD). Table 4 and Supplementary Material S4 present the results obtained from these experiments.

The model ultimately selected by the authors is the BERT-Multi+BiLSTM model with 284 hidden layers, an AdamW optimizer and a negative log-likelihood loss function. Several of the setups investigated for the BETO+BiLSTM and the RoBERTa BNE models present similar overall performance as the configuration proposed by the authors. The difference typically lies in class-specific performance, that is, lower performance in detecting stances against vaccination.

Discussion

The BERT-Multi+BiLSTM model obtained the best overall and class-specific performances (F1 = 0.86, MCC = 0.79, AUC = 0.96, and iAUC-against = 0.95, iAUC-None = 0.97, iAUC-In favor = 0.85). The language-specific BiLSTM models yielded similar average scores. In particular, the RoBERTa BNE+BiLSTM model (F1 = 0.83, MCC = 0.76, and AUC = 0.96) and the BETO+BiLSTM model (F1 = 0.86, MCC = 0.78, and AUC = 0.96) were used. The models using XLM-RoBERTa multilingual embedding showed poorer performance, notably in detecting stances in favor of vaccination. One of the models based on a traditional machine learning approach, that is, the TF-IDF+SVM model, also achieved high average performance (F1 = 0.86, MCC = 0.79, and AUC = 0.95) but showed slightly lower results when analyzing the interpolated precision-recall curves for the different classes.

The BERT-Multi+BILSTM model outperformed the other models in terms of per class prediction capacity. The main assumption is that language-specific embeddings do not outperform multilingual embeddings or TF-IDF features because of the context of the topic. The inherent context of BERT or RoBERTa embeddings is general, that is, BERT-Multi was trained on a Wikipedia corpus, BETO was trained on documents from Wikipedia and OPUS projects, and RoBERTa BNE was trained on documents from the National Library of Spain. These embeddings are not familiar with the slang present in the analyzed corpus, such as that commonly used on Twitter and, more specifically, during the pandemic (ie, words such as COVID-19, "plandemic," "covidiot," Sputnik, which means vaccine, booster dose, lockdowns,

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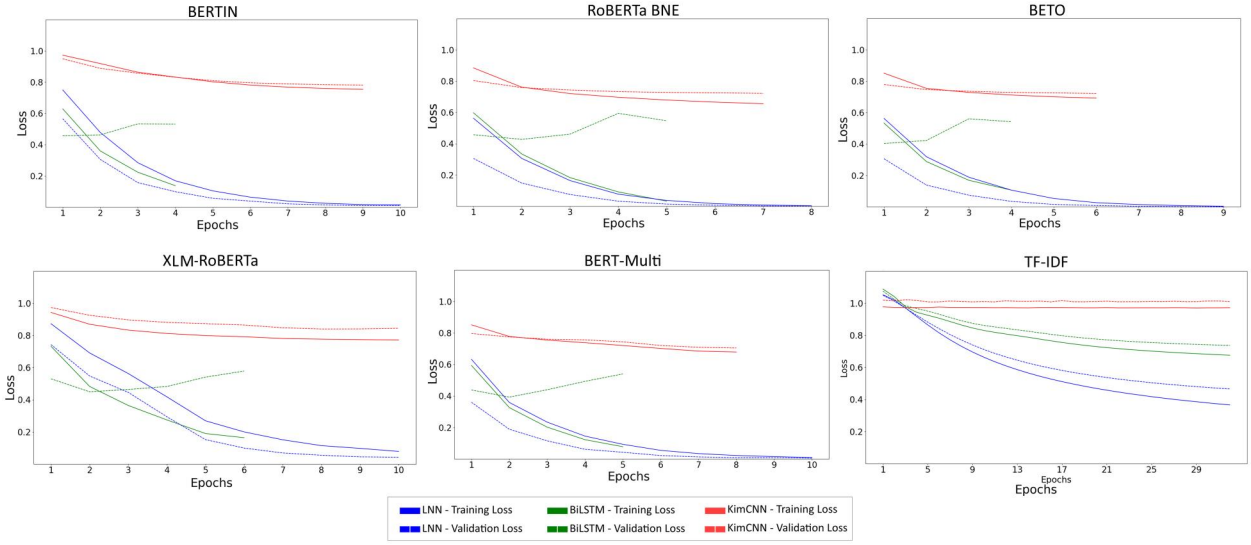

Figure 2. Training and validation loss during the training process. Early stopping was implemented for pretrained embeddings (patience = 2, minimum delta = 0.02, and maximum epochs = 15), and 30 epochs were considered for TF-IDF. Abreviation: TF-IDF, term frequency-inverse document frequency.

Table 3. Ablation study performance for architectures with TF-IDF with metrics of accuracy (Acc), precision (Prec), recall (Rec), F1 score (F1), Mathew correlation coefficient (MCC), area under the ROC curve (AUC), and area under the interpolated precision/recall curve (iAUC).

		TF-IDF+SVM									
Grams	Vector size	Acc	Prec	Rec	F1	MCC	AUC	iAUC	iAUC-Against	iAUC-None	iAUC-In favor
Uni	1024	0.86	0.87	0.83	0.84	0.75	0.96	0.92	0.91	0.96	0.87
Uni+Bi	1024	0.86	0.86	0.83	0.84	0.75	0.95	0.91	0.91	0.96	0.87
Uni+Bi+Tri	1024	0.86	0.86	0.82	0.84	0.75	0.95	0.91	0.91	0.96	0.87
Uni	512	0.85	0.86	0.82	0.84	0.74	0.95	0.91	0.90	0.96	0.86
Uni+Bi	512	0.85	0.86	0.82	0.83	0.74	0.95	0.90	0.89	0.95	0.86
Uni+Bi+Tri	512	0.85	0.85	0.81	0.83	0.74	0.95	0.90	0.89	0.95	0.86
Bi	512	0.69	0.73	0.54	0.57	0.43	0.81	0.71	0.55	0.87	0.71
Bi+Tri	512	0.70	0.73	0.54	0.57	0.44	0.80	0.70	0.55	0.86	0.69
Bi	1024	0.70	0.76	0.55	0.57	0.46	0.80	0.70	0.56	0.80	0.74
Bi+Tri	1024	0.71	0.76	0.56	0.59	0.47	0.79	0.68	0.53	0.78	0.73
Tri	1024	0.60	0.83	0.38	0.34	0.22	0.64	0.54	0.22	0.74	0.27
Tri	512	0.61	0.84	0.39	0.35	0.23	0.63	0.53	0.20	0.78	0.58

The experiments include different features in combination with the use of unigrams, bigrams, and trigrams.
Abbreviations: SVM, support vector machine; TF-IDF, term frequency-inverse document frequency.

"#yoMeVacuno," etc.). All the vocabularies were extended with the same list of topic-specific Spanish terms, including meaningful hashtags (see details in Supplementary Material S2). However, the extension was not comprehensive (ie, there will still be many unknown terms), some of the terms in the social postings are in English (eg, "leak," "lockdown," "plandemic," "now," or "fakenews"), and each embedding performs tokenization differently for unknown terms. For example, the BERT-Multi tokenizer breaks the word "lockdown" in 2 tokens, that is, "lock" and "down"; BERTIN breaks the term into "lock," "do," "w," and "n"; and BETO does not break it and uses the token unknown. A closer look into the results also points to a common practice during the pandemic, that is, the translation and publication of similar news in multiple languages, namely statistics reports with residual stances and messages defying lockdowns or vaccine efficacy. These posts do not contain a stance and use little language-dependent jargon and colloquial terminology; thus, multilingual and language-specific embeddings tend to grasp the message similarly.

BERT and RoBERTa embeddings trained on Twitter and COVID-19 conversations are likely to achieve better performance, but the tradeoff between performance improvement and the time and effort required to train such representations may not be practical. Training the embeddings over social media postings, preferably from different platforms, has the advantage that the models will be equipped to operate over social media postings broadly. However, the generalization to other biomedical domains still requires vocabulary augmentation, preferably resorting to semiautomatic means of characterizing domain lexical and linguistic idiosyncrasies.

Ethical considerations

The present research was conducted within the purview of acceptable privacy practices to avoid coercion and intrusive treatment. The present research involves publicly available Twitter data that were compiled following the platform's policies. The work focuses on developing a stance detection

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Table 4. Ablation study performance for the BERT-Multi+Bi-LSTM architecture with metrics of accuracy (Acc), precision (Prec), Recall (Rec), F1 score (F1), Mathew correlation coefficient (MCC), area under the ROC curve (AUC), and area under the interpolated precision/recall curve (iAUC).

			BERT-Multi+Bi-LSTM									
Hidden layers	Optimizer	LossFunction	Acc	Prec	Rec	F1	MCC	AUC	iAUC	iAUC-Against	iAUC-None	iAUC-In favor
284	AdamW	NLLLoss	0.88	0.87	0.86	0.87	0.80	0.97	0.93	0.96	0.98	0.83
384	AdamW	CrossEntropyLoss	0.89	0.87	0.87	0.87	0.80	0.97	0.94	0.96	0.98	0.88
284	AdamW	CrossEntropyLoss	0.89	0.87	0.88	0.88	0.81	0.97	0.93	0.96	0.97	0.88
384	AdamW	NLLLoss	0.87	0.85	0.85	0.85	0.77	0.97	0.93	0.95	0.97	0.85
184	AdamW	CrossEntropyLoss	0.89	0.88	0.88	0.88	0.81	0.96	0.93	0.96	0.98	0.87
184	AdamW	NLLLoss	0.87	0.85	0.84	0.85	0.78	0.96	0.92	0.94	0.97	0.84
284	SGD	CrossEntropyLoss	0.28	0.41	0.35	0.28	0.05	0.57	0.40	0.22	0.68	0.30
384	SGD	CrossEntropyLoss	0.36	0.35	0.35	0.31	0.04	0.53	0.36	0.19	0.60	0.27
384	Adagrad	NLLLoss	0.57	0.19	0.33	0.24	0.00	0.52	0.35	0.20	0.59	0.26
184	SGD	CrossEntropyLoss	0.39	0.33	0.33	0.33	−0.01	0.51	0.35	0.24	0.58	0.21
184	SGD	NLLLoss	0.45	0.34	0.34	0.30	0.02	0.50	0.34	0.20	0.58	0.23
184	Adagrad	CrossEntropyLoss	0.57	0.19	0.33	0.24	0.00	0.50	0.33	0.18	0.58	0.22
184	Adagrad	NLLLoss	0.57	0.19	0.33	0.24	0.00	0.50	0.34	0.20	0.59	0.23
284	Adagrad	CrossEntropyLoss	0.57	0.19	0.33	0.24	0.00	0.50	0.34	0.23	0.58	0.21
384	Adagrad	CrossEntropyLoss	0.57	0.19	0.33	0.24	0.00	0.49	0.33	0.19	0.57	0.22
284	Adagrad	NLLLoss	0.57	0.19	0.33	0.24	0.00	0.49	0.33	0.21	0.56	0.23
384	SGD	NLLLoss	0.36	0.33	0.32	0.32	0.01	0.48	0.33	0.16	0.60	0.22
284	SGD	NLLLoss	0.28	0.32	0.32	0.27	−0.02	0.48	0.32	0.18	0.53	0.24

Parts of the architecture, such as the hidden layers, optimizers, and loss functions, were changed.
Abbreviations: NLLLoss, negative log-likelihood loss; SGD, stochastic gradient descent.

model for screening public health concerns and does not make any claims related to the health topic under analysis.

Bias

The free Twitter API supports tweet retrieval, making data collection susceptible to demographic and time-specific biases, which can cause latent problems, especially when inferences are incorporated into real-life situations. We attempt to reduce these existing biases in the dataset by stratifying the split of tweets into training, validation, and test sets.

Stance annotation of the dataset was performed following existing guidelines and consensus classes. However, the stance is subjective in nature, and social posts often contain idiosyncrasies that may result in varying interpretations of tweets across individuals. Therefore, the interannotator agreement was not perfect, and many tweets were discarded because of the inability to reach an agreement. Another common feature of corpora based on social media is that a small number of users often generate a large proportion of posts, and vice versa. As tweets from the same author might contain specific information about the user, such as individual writing style, vocabulary or other communicative behavior, models could learn those specific features and thus overfit the characteristics of some specific users. Corpus extension via semisupervised learning could be an option, but a well-reasoned selection policy (most likely topic-oriented) should be implemented to guarantee adequate classifier properties.

Practical applicability and future work

We study Twitter data in a purely observational and nonintrusive manner without any intervention or alteration in the user experience. Stance detectors aim to provide an assessment of individuals' positions on relevant topics; thus, they can aid in the design of awareness campaigns, misinformation debunking and other public health applications. Evaluating multilingual and language-specific stance detectors across social media platforms paves the way for our future work. Other datasets from the same platform (ie, inspecting the dialogues in other Twitter communities) and other social media platforms (eg, Instagram and Reddit) are needed to grasp the functioning of each social media platform and the dynamics surrounding given topics and certain user communities comprehensively. Some communities and topics of conversation are naturally depicted by strong, sometimes highly volatile, stances, whereas others conform to milder, more conservative stance expressions.

Conclusions

Health topics are a main, timely target of social analyses, and the implementation of specialized stance detectors for main languages other than English is of practical relevance. The present study investigates stance detection on social media posts in the Spanish language. The results showed that language-specific embeddings do not outperform the BERT multilingual embedding. These observations are relevant because, in the context of a worldwide health crisis, it is critical to reach the public as fast and accurately as possible, knowing that, initially, people's stance can be monitored with preexisting, multilingual resources. This head start is valuable for monitoring user communities while developing specialized resources that are able to provide a deeper understanding of people's opinions and feelings as the situation progresses.

The results are also relevant in terms of generalization to other biomedical domains or topics of conversation. Specifically, the addition of topic-specific terminology and fine-tuning of the embeddings were needed to better capture the context of the social postings. Future research will be devoted to semiautomated analysis of the impact of topic-specific terminology and necessary tuning of embeddings.

Author contributions

Guillermo Blanco and Rubén Yáñez Martínez: Data curation, Investigation, Software, Resources, Writing – original draft.

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Anália Lourenço: Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing.

Supplementary material

Supplementary material is available at *JAMIA Open* online.

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Conflicts of interest

None declared.

Data availability

The data and the source code underlying this article are available at https://github.com/gbgonzalez/stance_detection_sp_vaccines.

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